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Title: Management of children with pulmonary hypertension

1. Introduction:

This protocol aims at organising the care of children with pulmonary hypertension. The etiology of pulmonary hypertension is variable and hence it has been classified in to five different classes. The management of these children depends on the underlying cause and often requires a multi-disciplinary approach. Initiating pulmonary vasodilator treatment ideally requires cardiac catheterization for acute vasodilator testing and monitoring is usually guided by serial echocardiograms both of which are under the expertise of pediatric cardiology.

2.Classification of pulmonary hypertension

Box 35.1 Updated Clinical Classification of Pulmonary Hypertension

- Pulmonary arterial hypertension
 - Idiopathic PAH
 - Heritable PAH
 - BMPR2
 - ALK 1, ENG, **SMAD9**, **CAV1**, **KCNK3**
 - Unknown
 - Drug and toxin induced
 - Associated with:
 - Connective tissue disease
 - HIV infection
 - Portal hypertension
 - Congenital heart diseases
 - Schistosomiasis
- 1' Pulmonary veno-occlusive disease and/or Pulmonary capillary hemangiomatosis**
- 1" Persistent Pulmonary Hypertension of the Newborn (PPHN)**
- Pulmonary hypertension due to left heart disease
 - Left ventricular systolic dysfunction
 - Left ventricular diastolic dysfunction
 - Valvular disease
 - Congenital/acquired left heart inflow/outflow tract obstruction and congenital cardiomyopathies**
- Pulmonary hypertension due to lung diseases and/or hypoxia
 - Chronic obstructive pulmonary disease
 - Interstitial lung disease
 - Other pulmonary diseases with mixed restrictive and obstructive pattern
 - Sleep-disordered breathing
 - Alveolar hypoventilation disorders
 - Chronic exposure to high altitude
 - Developmental lung diseases
- CTEPH
- Pulmonary hypertension with unclear multifactorial mechanisms
 - Hematologic disorders: **chronic hemolytic anemia**, myeloproliferative disorders, splenectomy
 - Systemic disorders: sarcoidosis, pulmonary histiocytosis, lymphangioleiomyomatosis
 - Metabolic disorders: glycogen storage disease, Gaucher disease, thyroid disorders
 - Others: tumoral obstruction, fibrosing mediastinitis, chronic renal failure, **segmental PH**

BMPR, Bone morphogenic protein receptor type II; *CAV1*, caveolin 1; *CTEPH*, chronic thromboembolic pulmonary hypertension; *ENG*, endoglin; *PAH*, pulmonary arterial hypertension; *PH*, pulmonary hypertension.

Fifth WSPH Nice 2013. Main modifications to the previous Dana Point classification are in **bold**.

Simonneau G, Gatzoulis MA, Adatia I, et al. Updated clinical classification of pulmonary hypertension. J Am Coll Cardiol. 2013;62 (25 suppl):D34-D41.

a. Referral for suspected pulmonary hypertension should be directed to cardiology first. If an underlying cardiac cause is excluded then all patients should be referred to pulmonology to rule out an underlying lung disease. After a thorough assessment by pulmonology, if an underlying lung disease or sleep disordered breathing is excluded, the patient to be referred back to cardiology for further follow up and management. Evaluation of an underlying connective tissue, metabolic or genetic disorder to be considered based on individualised case assessment (see algorithm below).

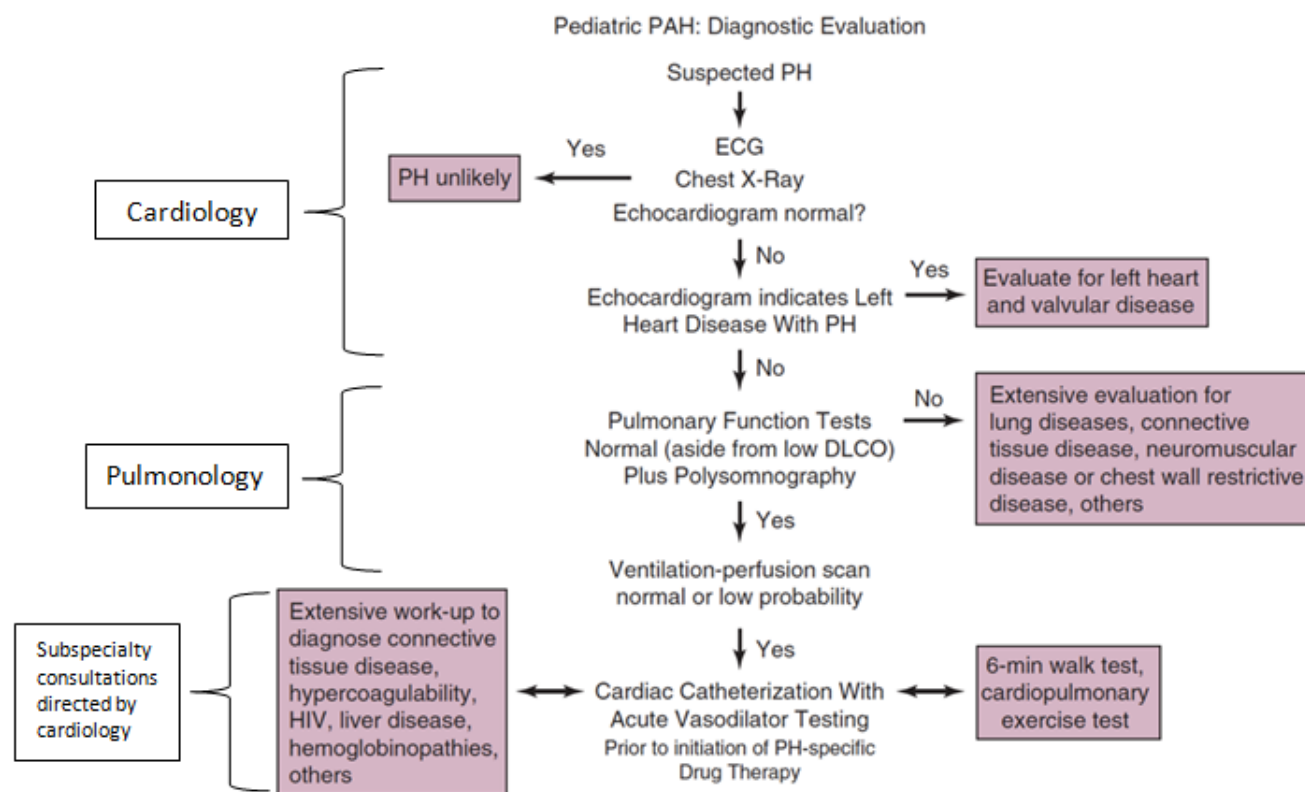


Fig. 35.3 Algorithm illustrating general diagnostic workup for pediatric pulmonary arterial hypertension. *DLCO*, Diffusing capacity of the lungs for carbon monoxide; *ECG*, electrocardiogram; *HIV*, human immunodeficiency virus; *PH*, pulmonary hypertension. (From Abman SH, Hansmann G, Archer SL, et al. Pediatric pulmonary hypertension: guidelines from the American Heart Association and American Thoracic Society. *Circulation*. 2015;132(21):2037-2099.)

4. Follow up of patients with pulmonary hypertension

a. Patient with pulmonary hypertension secondary to underlying lung disease (Class 3) will be primarily follow by pulmonology team for the treatment of underlying lung disease. However, the monitoring of pulmonary hypertension progression by echocardiogram and catheterization when needed and the titration of targeted pulmonary hypertension treatment will be by through regular OPD follow up with cardiology.

b. Patient with pulmonary hypertension secondary to underlying systemic disease (Class 5) will be primarily follow by the respective team (depending on the underlying disease). However, the monitoring of pulmonary hypertension progression through echocardiogram and the titration of treatment will be by through regular OPD follow up with cardiology.

c. All other classes of pulmonary hypertension will be followed and managed by the cardiology team.

5. Admission of children with pulmonary hypertension

a. When a patient with pulmonary hypertension secondary to underlying lung disease (Class 3) requires admission, he/she to be admitted under pulmonology. The pulmonology team will assess the need for a cardiology consultation to advice regarding the titration of medical therapy.

b. Patient with pulmonary hypertension secondary to underlying systemic disease (Class 5) will be admitted under respective team (depending on the underlying disease). Cardiology consultation will be offered if indicated.

c. Children with other classes of pulmonary hypertension admitted with pulmonary hypertension exacerbations should be admitted under cardiology.

d. Children admitted with moderate to severe pulmonary hypertension should be closely monitored and managed in medical /cardiac HD or ICU based on etiology.

6. Management of children with pulmonary hypertension admitted to intensive care unit

b. For children with other classes of pulmonary hypertension who are admitted to PICU/HD with a concern about worsening pulmonary hypertension, a cardiology referral is indicated.

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